

Tut - 7 Intro to filters

1. Determine a closed-form expression for the frequency response of the LTI DT system characterized by an impulse response

$$g(n) = h(n) * h(n) * h(n),$$

$$\text{where } h(n) = \delta(n) - \alpha \delta(n-R) \text{ where } |\alpha| < 1.$$

2. S.T. Group delay $\tau(\omega)$ of an LTI DT system characterized by a freq. response $H(e^{j\omega})$ can be expressed as

$$\tau(\omega) = \text{Re} \left\{ \frac{j \frac{dH(e^{j\omega})}{d\omega}}{H(e^{j\omega})} \right\}.$$

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